

Predicting 30-Day Readmission in Hospitalised Patients with Type 2 Diabetes Using Machine Learning

Abstract

Unplanned readmission within 30 days of discharge is a widely used indicator of care quality and a substantial driver of hospital cost. We developed and evaluated machine learning models to predict 30-day readmission among inpatients with type 2 diabetes mellitus, using routinely collected clinical and administrative data available at the point of discharge.

Methods

Study population and data source

We retrospectively analysed the electronic health records of adult inpatients discharged from a tertiary teaching hospital. Patients younger than 20 years and those who died during the index admission were excluded.

Predictor variables

Ten variables available at discharge were used as model inputs:

Variable	Type	Description
age	numerical	Patient age in years at admission
gender	categorical	Biological sex, recorded as M or F
bmi	numerical	Body mass index at admission (kg/m ²)
hba1c	numerical	Glycated haemoglobin (%) measured during the index admission
systolic_blood_pressure	numerical	Systolic blood pressure at admission (mmHg)
length_of_stay	numerical	Duration of the index admission, in days
admission_date	date	Date of admission for the index episode
diabetes_diagnosis	categorical	Whether type 2 diabetes was a recorded diagnosis
pain_score	numerical	Numeric rating scale for pain at admission (0-10)
charlson_index	numerical	Charlson comorbidity index score

The outcome variable was readmission_30d, a binary indicator of unplanned readmission to any department within 30 days of discharge.

Preprocessing

Missing values in continuous predictors were imputed with the column mean. All continuous predictors were standardised to zero mean and unit variance. Categorical predictors were one-hot encoded.

Class imbalance

Because 30-day readmission is comparatively rare in this cohort, we applied SMOTE to the training partition only, using five nearest neighbours. Resampling was never applied to the validation or test partitions.

Models

We compared logistic regression as a baseline against Random Forest and XGBoost.

Validation and metrics

Models were assessed with stratified 10-fold cross-validation. We report balanced accuracy, AUC, AUPRC, F1, recall and specificity. Because the outcome is imbalanced, AUPRC was treated as the primary metric.

Results

XGBoost achieved the highest AUC, followed by Random Forest. Length of stay, HbA1c and the Charlson comorbidity index were the three most influential predictors in the final model.

Conclusion

Routinely collected discharge-time variables can identify patients at elevated risk of 30-day readmission with useful discrimination, supporting targeted follow-up for high-risk patients.